

Chapter 6

Special Relativity

6.1 Basic concepts

6.1.1 Comparison with Newtonian dynamics

Three basic assumptions of Newtonian dynamics are:

1. There is a special class of reference frames, called inertial frames: an inertial frame is one in which Newton's first law holds.
2. Any two inertial frames are related by a Galilean transformation.
- 2' Time is absolute.
3. Newton's second law (with Galilean invariant forces, including the universal law of gravitation) holds in any inertial frame. This is sometimes called the *principle of Galilean relativity*.

The corresponding assumptions from Special Relativity are:

1. Same as the Newtonian assumption.
2. Any two inertial frames are related by a Lorentz transformation¹.
- 2' The speed of light is absolute.
3. Newton's second law, with momentum suitably defined, (with Lorentz invariant forces, including the Lorentz force²) holds all inertial frames. This is sometimes called the *principle of special relativity*.³

6.1.2 Consequences for Special Relativity

Some consequent differences between Newtonian Dynamics and Special Relativity are as follows.

- Newtonian absolute time is replaced by absolute speed of light: in Special Relativity, the speed of light in vacuo is c (about 3×10^8 metres per second) for all observers.
- In Special Relativity, the notion of simultaneity is frame-dependent. This is a simple consequence of the fact that time is no longer absolute: each observer has his or own time and so observers will not agree on which events are simultaneous.
- In Special Relativity, lengths are contracted and time is dilated⁴ in frames moving relative to the observer. This does not follow obviously from the assumptions of the previous section; it will be derived once we have an algebraic form for Lorentz transformations.

¹See section 6.3 for the definition of a Lorentz transformation.

²Note to self: did I ever mention that the Lorentz force is not invariant under Galilean transformations? (It is approximately invariant for small speeds.) If not, it was a bit naughty to stick the Lorentz force on the right hand side of N2.

³The 'special' refers to the fact that only inertial frames are considered; the principle of general relativity is (roughly) that the laws of physics are the same in all frames. It was (I think) Max Planck who used the name 'special relativity'; Einstein himself, in his 1905 paper, did not call his theory anything particular: his paper was entitled 'On the electrodynamics of moving bodies'.

⁴clocks tick slower

- There is a different velocity addition law in Special Relativity which means that the speed of light cannot be attained by massive particles. Clearly, if the speed of light is the same in all frames, we cannot use the Galilean formula $c' = c + v$ for the velocity c' of light in a frame that is moving at velocity v with respect to the original frame.
- New definitions for energy and momentum; for example ' $E = mc^2$ ' (though we will never use the definition of energy in this form).

6.1.3 The need for Special Relativity

The invariance of the speed of light that led Einstein to propose a theory invariant under Lorentz transformations instead of Galilean transformations is both an experimental and a theoretical result.

The most significant experiment was performed by Michelson and Morley (actually a series of experiments beginning in 1887), in which they tried to detect the motion of the Earth through the ether⁵. However, they failed to detect any difference in light speed in orthogonal directions even when they took the apparatus up a high mountain (in case the ether was stuck to the Earth at low altitudes). Fitzgerald and Lorentz independently proposed that the effect of the ether on a moving body was to exert a force that compressed the body by a factor $(1 - v^2/c^2)^{\frac{1}{2}}$. It was later shown that the idea of the contraction being caused by a physical force was untenable.

On the theoretical side, Maxwell's equations for electromagnetic fields showed that the fields propagated at a speed almost exactly equal to the measured speed of light. In 1903, Lorentz showed that Maxwell's equations are invariant under what are now called *Lorentz transformations*⁶, implying that the speed at which the waves travel would be the same in all frames related by Lorentz transformations.

In 1905, from a very different starting point, Einstein showed that if the speed of light is the same in all frames, then the frames are related by Lorentz transformations.

6.2 Space-time diagrams

As in Newtonian dynamics, it is often convenient to exhibit information by means of a diagram. The usual convention in relativity is to have the time axis vertical and the space axis or axes horizontal; this is the other way round from the usual convention in A-level mechanics (for example). Thus a particle at rest moves along a vertical line. In particular, an observer at rest at the spatial origin $x = 0$ moves up the time axis. In this case, we say that the axes correspond to the *observer's frame* or the *observer's frame of reference*; the observer's frame is the set of axes for which the observer moves along the trajectory $x = 0$.

The trajectory of a particle or observer in space-time is called a *world line*. The gradient of the world line of a physical particle must be steeper than that of a light ray because it cannot move faster than the speed of light.⁷ In two dimensions (one time, one space), the pair of light trajectories, one moving to the left and one to the right, through a point P is called the *light cone*. In three dimensions (one time, two space), the light cone through P would look like a double cone, one cone representing light spreading out from P and the other representing light converging on P . In four dimensions (one time, three space) the light cone would consist of one set of expanding spherical surfaces corresponding to a flash of light spreading out from P , and one set of contracting spherical surfaces representing light arriving at P .⁸

The units on the axes of a space-time diagram are chosen to have the same dimension by using ct rather than t on the time axis, so that one unit up the ct axis represent a time taken by light to travel one unit of distance. This means that the light cone, represented by dashed lines, is at 45° to the axes.

Each point in a space-time diagram represents an *event*.⁹ If the gradient of the straight line joining two points in a space-time diagram is greater than one in magnitude (i.e. it is steeper than

⁵It was thought at that time that light must propagate through some all pervading medium, which was called the ether. If this were the case, and the Earth was moving through the ether, light should travel at different speeds in orthogonal directions: compare with a swimmer in a river swimming either with the current or across the current.

⁶Lorentz discovered these transformations after playing around with the effect of different transformations on the Maxwell equations which were known not to be invariant under Galilean transformations.

⁷If we assume that there is an inertial frame in which the observer is at rest, and that inertial frames are related by Lorentz transformations, then there is no inertial frame in which an observer moves as fast as the speed of light. This will become clearer in the next section.

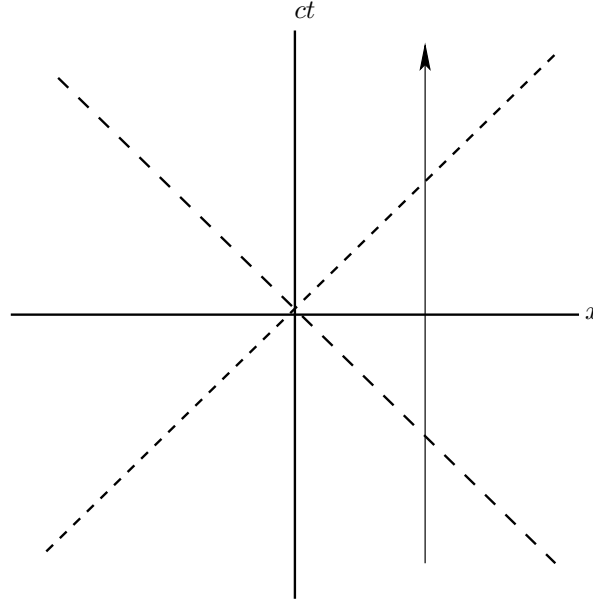
⁸These light cones are difficult to picture, being three-dimensional surfaces embedded in four-dimensional space-time; suppressing one space dimension results in the easily visualised double cones.

⁹We will not always make a distinction between the event and the point representing it.

a light ray), then the corresponding events are said to be *causally related*; this means that one event could influence the other by sending a signal which did not have to travel faster than the speed of light. If event E_1 occurs within the light cone of event E_2 , then E_1 and E_2 are causally related.

If events represented by points A and B in a space-time diagram are causally related, the straight line joining A and B in the space-time diagram is said to be *time-like*; if A lies on the light cone through B , the line is said to be *null* or *light-like*; otherwise, A and B are connected by a *space-like* line.

The *rest frame* of an observer is a frame in which the observer's x coordinate is constant. The world line is represented on a space-time diagram by a vertical line; normally one refers to the rest frame as being the frame in which the observer is at $x = 0$, so his or her world line is the ct axis. The *steeper* the gradient of the world line of an observer, the *slower* the observer is moving in that frame.



A space-time diagram. The x -axis is horizontal and the ct axis is vertical. The vertical line with an arrow is the world line of an observer at rest in these axes.

6.3 Lorentz transformations (one space dimension case)

6.3.1 Definition

In two dimensions (one space, one time) we define a Lorentz transformation relating a frame S with coordinates (ct, x) and a frame S' with coordinates (ct', x') moving with velocity v relative to S by

$$x' = \gamma(x - vt) \quad (6.1)$$

$$t' = \gamma(t - vx/c^2) \quad (6.2)$$

where

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}. \quad (6.3)$$

Note that $\gamma \geq 1$ (and is real, since we assume that $|v/c| \leq 1$). If $v/c \ll 1$ we retrieve the Newtonian limit which is the Galilean transformation $x' = x - vt$, $t' = t$.

Since the transformations are linear, they also relate displacements between two space-time events, which may be infinitesimally close:

$$\Delta x' = \gamma(\Delta x - v\Delta t) \quad (6.4)$$

$$\Delta t' = \gamma(\Delta t - v\Delta x/c^2) \quad (6.5)$$

or

$$dx' = \gamma(dx - vdt) \quad (6.6)$$

$$dt' = \gamma(dt - vdx/c^2) \quad (6.7)$$

If we take as an axiom in Special Relativity that inertial frames are related by Lorentz transformations instead of by the Newtonian Galilean transformations, we find that the speed of light is the same in all inertial frames: if $x = ct$ then

$$\begin{aligned}
 x' &= \gamma(x - vt) && \text{(by the definition of the Lorentz transformation (6.1))} \\
 &= \gamma(ct - vt) && \text{(since } x = ct) \\
 &= \gamma(ct - vx/c) && \text{(since } t = x/c) \\
 &= ct' && \text{(by the definition of the Lorentz transformation (6.2))}
 \end{aligned}$$

Alternatively, we could define Special Relativity by the requirement that the transformations between inertial frames are such that:

- Newton's first law holds in all inertial frames, so that a particle moving with constant velocity in one inertial frame also moves with constant velocity in any other inertial frame (or, to put it geometrically, straight lines in space-time diagrams map to straight lines);
- the speed of light is the same in all inertial frames (or, to put it geometrically, the straight lines in space-time diagrams with gradient $\pm c$ map to straight lines with the same gradient).

We can then show that inertial frames are related by Lorentz transformations. The proof is not very interesting.¹⁰

It is helpful to picture the effect of a Lorentz transformation in a space-time diagram. The important thing to notice is that in the space-time diagram corresponding to the frame S , the axes of a frame S' moving with speed v relative to S are inclined symmetrically to the light cone. We can see this from the transformations (6.1) and (6.2). The time axis for S' has equation $x' = 0$ which corresponds to

$$x = vt = \frac{v}{c}(ct)$$

in S (not surprisingly: the frame is moving with speed v relative to S). The space axis for S' has equation $t' = 0$ which corresponds to

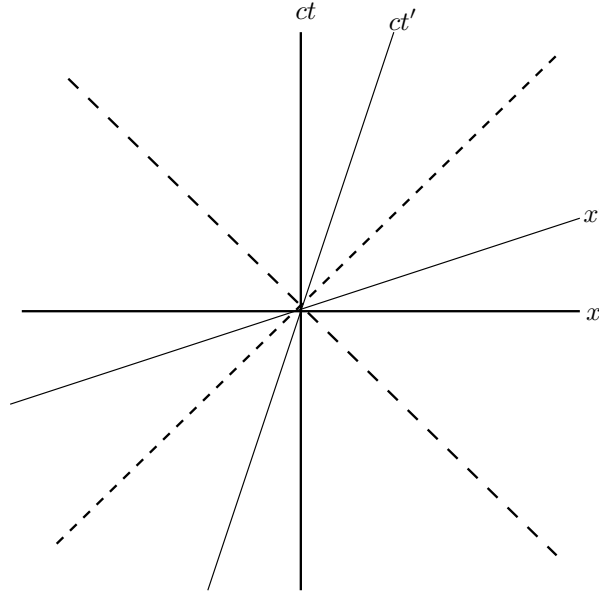
$$t = vx/c^2, \quad \text{i.e.} \quad x = \frac{c}{v}(ct)$$

in S . Since the gradients of these lines are reciprocal, the two lines are equally inclined to the t -axis and the x -axis respectively, as shown below.

¹⁰If the transformation from frame S to frame S' is $x' = f(x, t)$, $t' = g(x, t)$ a particle moving with speed v in S moves with speed v' in S' , where

$$v' = \frac{f_x v + f_t}{g_x v + g_t},$$

where the subscripts denote partial differentiation. Since v' is constant whenever v is constant, and remembering that f and g do not depend on v , one can argue or prove that f and g are linear in both x and t . Finally, using that conditions that $v' = c$ when $v = c$, and $v' = -c$ when $v = -c$ nails the Lorentz transformation (up to an overall multiple which corresponds to a change of units of both x and t . The argument is a bit finicky, because it is too glib to say that ' f and g do not depend on v '. It is true, of course; but the *arguments* of these functions can (and will), so there is still quite a lot of work to do.)



6.3.2 Matrix representation

Sometimes it is helpful to write the Lorentz transformation in matrix form:

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \gamma & -v\gamma/c \\ -v\gamma/c & \gamma \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}. \quad (6.8)$$

The matrix representation of the two-dimensional Lorentz transformation works out neatly if the transformation is parameterised by a hyperbolic angle β , instead of by v/c , where β is defined by

$$\tanh \beta = \frac{v}{c}. \quad (6.9)$$

In terms of this new parameter,

$$\gamma = \frac{1}{\sqrt{1 - \tanh^2 \beta}} = \cosh \beta$$

so we can write the matrix (6.8) as

$$\begin{pmatrix} \cosh \beta & -\sinh \beta \\ -\sinh \beta & \cosh \beta \end{pmatrix} \equiv L(\beta). \quad (6.10)$$

Note the similarity to a rotation matrix. In fact (and this is a spoiler for section 6.6), the exact correspondence is that the Lorentz transformation preserves the matrix

$$\eta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (6.11)$$

in the sense

$$L^T \eta L = \eta \quad (6.12)$$

as can easily be verified directly, and a rotation matrix preserves the unit matrix I :

$$R^T I R = I.$$

We see immediately from this matrix form (6.10) that Lorentz transformations form a group, the multiplication law being

$$L(\beta_1)L(\beta_2) = L(\beta_1 + \beta_2).$$

In particular, $(L(\beta))^{-1} = L(-\beta)$ which is what one would expect on physical grounds.¹¹

¹¹You ought to be able to return from the moving frame S' to your original frame S , which is moving with velocity $-v$ relative to your transformation, by applying the inverse transformation.

6.3.3 Simultaneity

In Newtonian dynamics, where time is absolute, all observers agree on the set of events that are simultaneous with a given event. Clearly, they will not agree in Special Relativity because the transformation law (6.2) means that time is different in different frames.¹² To an observer stationary in S , all events with

$$t = \text{constant}$$

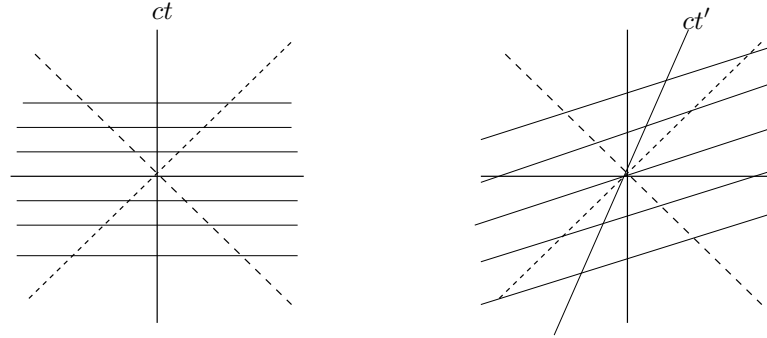
are simultaneous. Similarly, to an observer stationary in S' , the all events with

$$t' = \text{constant}$$

but this corresponds to

$$t - vx/c^2 = \text{constant}$$

in S .



Lines of simultaneity in S (horizontal, in the diagram on the left) and in S' (slanted, in the diagram on the right).

Note that, given an event E outside the light cone of the event $(0,0)$, there exists a Lorentz transformation to a new frame in which these two events are simultaneous, but if the event E lies inside the light cone of the event $(0,0)$, there is no Lorentz transformation to a new frame in which these two events are simultaneous. It therefore makes sense to say that events within the light cone of $(0,0)$ with $t > 0$ (the *future* light cone) are to the future of $(0,0)$ and events within the light cone of $(0,0)$ with $t < 0$ (the *past* light cone) are to the past of $(0,0)$; this statement is invariant under Lorentz transformations (there is no frame, related to the original frame by a Lorentz transformation, in which the statement is false). It follows that the event at the origin can be influenced by events within or on the past light cone but not by other events (assuming of course that no signals can travel fast than the speed of light).

6.4 Time dilation and length contraction

These two phenomena are closely related: the same observations can often be explained in one frame in terms of time dilation and in another frame in terms of length contraction. It is helpful, before defining exactly what these terms mean and doing the mathematics, to give an example.

6.4.1 Meson decay

A muon is a charged particle that decays into an electron or positron, a neutrino and an anti-neutrino:

$$\mu^+ \rightarrow e^+ + n_1 + n_2 \quad \text{or} \quad \mu^- \rightarrow e^- + n_1 + n_2$$

Muons occur in cosmic rays travelling through the atmosphere at speeds very close to that of light.¹³ In 1941, Rossi and Hall measured the flux of muons in a laboratory at 6300 feet above sea level (top of Mount Washington) and in a laboratory at 2000 feet above sea level (bottom of Mount Washington).

At the top they measured 550 muons per hour. At the foot (simultaneously) they measured 422 muons per hour. The half-life of the muon is 1.56 microseconds.

¹²They don't have to argue over it: all they have to do is respect the other observer's point of view and then there is no inconsistency, though you wouldn't think so from the spats that occur in some web sites.

¹³They are created by collisions between protons in cosmic rays with atoms of air in the upper atmosphere.

From this information, one can calculate how long the muons spend travelling between the two laboratories and hence the speed of the muons *and the speed turns out to be much faster than the speed of light*. The muons travel a distance D at speed v taking time $T = D/v$; during this time the number $N(T)$ of muons remaining is given by

$$N(T) = N(0) \left(\frac{1}{2}\right)^{(T/T_{\text{half}})}.$$

Thus

$$v = \frac{D}{T} = \frac{D \log \frac{1}{2}}{T_{\text{half}} \log (N(T)/N(0))} = \frac{4300 \log \frac{1}{2}}{1.56 \log (422/550)} = 7212$$

in units of feet per microsecond. The speed of light in these units is about 1000.

To put it another way, the observed flux of muons at the lower laboratory is far too high for particles covering the distance at less than the speed of light: many more should have decayed in the travel time.

What is the explanation? As we will see, it depends on whether we work in the rest frame of the muon or the rest frame of the laboratory, the two being in relative motion at close to the speed of light.

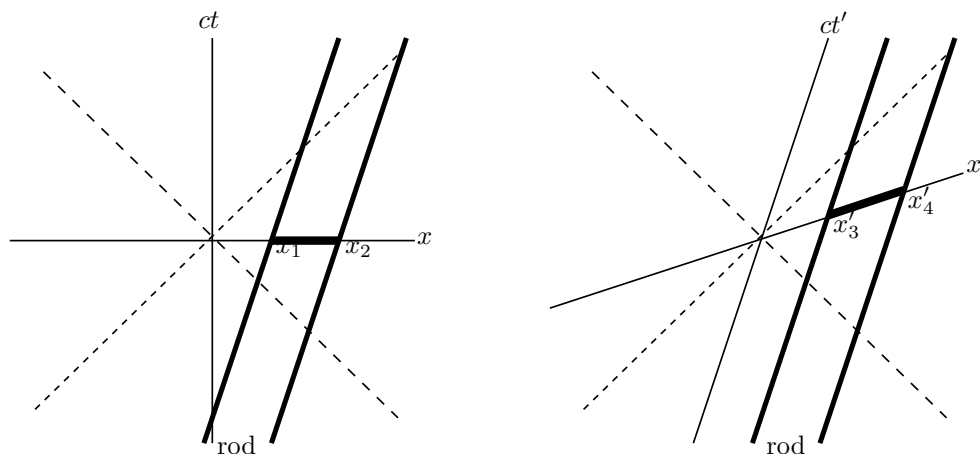
In the rest frame of the laboratory, the explanation is time dilation: time in the moving frame is dilated relative to time in the rest frame, which means that clocks are ticking slower, by a factor of γ , in the moving frame. Suppose the travel time is T seconds as measured (by distance/speed) in the laboratory frame. Then in this interval, only T/γ seconds have elapsed in the moving muon frame, so far fewer muons will decay, corresponding to a half-life of 1.56γ microseconds.

But how can this be explained in the muon frame, where the half-life is 1.56 microseconds? The explanation now is length contraction. In the rest frame of the muon, Mount Washington, which is zooming towards the muon at high speed, is only $6000/\gamma$ feet high, because lengths of moving rulers are contracted. Thus the time taken to cover this contracted distance is short: only T/γ seconds. There is little time for the muons to decay.

6.4.2 Length contraction

We consider a rod of length L' (in its own rest frame). It is moving (along its length, like a javelin) at velocity v relative to the lab frame.

The space-time diagrams below show the world lines of the two ends of the rod, in the lab frame (left hand diagram) and in the rest frame of the rod (right hand diagram).



The event with coordinates $(0, x_1)$ and $(0, x_2)$ in S are simultaneous in S and the events with coordinates $(0, x'_3)$ and $(0, x'_4)$ in S' are simultaneous in S' .¹⁴

What does it mean to say that the rod has length L in S ? Clearly, you have to contrive to measure the position of each end of the rod at the same time, and then subtract the x coordinates: that will give you the length. Thus the length L in S of the moving rod is given by

$$L = x_2 - x_1$$

¹⁴You can save a bit of writing by choosing axes such that $x_1 = x'_3 = 0$; it is just a translation of the origins of the two frames.

and the length L' of the rod in its rest frame S' is given by

$$L' = x'_3 - x'_4.$$

Our task is to relate L and L' .

This is only a matter of coordinate geometry.¹⁵

The equation in S' of the world line of the leading end of the rod is

$$x' = x'_4$$

and the equation of the trailing end is

$$x' = x'_3.$$

We can use the Lorentz transformations (6.1) and (6.2) to transform these lines into S :

$$\gamma(x - vt) = x'_4$$

and

$$\gamma(x - vt) = x'_3.$$

These lines meet the x -axis ($t = 0$) at $x = x_2$ and $x = x_1$ respectively, so

$$\gamma x_1 = x'_3 \quad \text{and} \quad \gamma x_2 = x'_4.$$

Putting these together;

$$L = x_2 - x_1 = \frac{x'_3 - x'_4}{\gamma} = \frac{L'}{\gamma}$$

which represents a contraction since $\gamma \geq 1$; the moving rod is shorter in the laboratory frame than if it were at rest in the laboratory frame.¹⁶

But hold on! If we look at the lengths of the rod, marked by the short heavy lines in the two diagrams, it seems that we have got it the wrong way round: the $(x'_4 - x'_3)$ definitely looks longer than $(x_2 - x_1)$. This is a trap: lengths in space-time diagrams are not like lengths in the more familiar x - y plane and we must rely on our calculations.¹⁷

6.4.3 Time dilation

We consider a clock moving at velocity v with respect to the laboratory frame. It ticks at times t'_1 and t'_2 , measured in its rest frame S' , where $t'_2 - t'_1 = \Delta t'$. These events (the ticks) have coordinates $(ct'_1, 0)$ and $(ct'_2, 0)$ in S' ,

Let the coordinates of the ticks in S be (ct_1, x_1) and (ct_2, x_2) . Then using the Lorentz transformation (6.2) with v replaced by $-v$ (because the lab frame moves with velocity $-v$ with respect to the clock's frame)

$$t_1 = \gamma t'_1 \quad \text{and} \quad t_2 = \gamma t'_2. \quad (6.13)$$

and Δt , the time interval between ticks in S , is given by

$$\Delta t = \gamma \Delta t' \geq \Delta t'.$$

In this interval, an identical clock at rest in S would have ticked more than once.

The precise situation described above concerns a lab frame S and two events in another (moving) frame S' . The time interval between the two events is longer in the lab frame than in the moving frame. This is exactly the situation in the meson decay experiment. At the first event, there is a flux of N mesons per second and at the second event there is a flux of $N/2$ mesons per second; the time interval between these events is the half-life. The time between these two events in the lab frame is longer (the half-life is longer in the lab frame than in the rest frame of the

¹⁵There are lots of ways of getting the required answer. Sometimes, you find at the end that you have a factor $\gamma(1 - v^2/c^2)$, which simplifies to γ^{-1} ; this usually indicates that there was a quicker way, for example, transforming from S to S' rather than from S' to S .

¹⁶Note that we say 'is shorter' not 'appears shorter'. It is important not to confuse the fact of being shorter with the appearance of being shorter: the appearance might be the result of taking a photograph, for example, which would be affected by the time delay of light travelling to the camera.

¹⁷To anticipate section 6.5: lengths are shorter the closer the inclination to the 45° of the null cone. This is because instead of the Euclidean norm (Pythagoras), one must use the norm $|(ct, x)| = (c^2 t^2 - x^2)^{1/2}$.

meson), which is why fewer decay in the transit time than would have been expected in Newtonian dynamics. This is *time dilation*.¹⁸

Of course, one can present this result from the point of view of the observer in S' , who sees S as the moving frame. Such an observer would notice that the time interval between two events in his or her rest frame (S') is shorter than in any other frame.

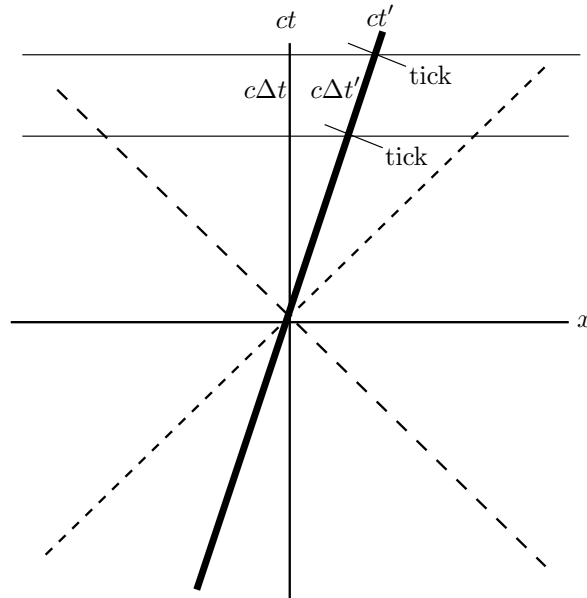
It is instructive to see what would have happened if we had woodenly used the forward transformation (6.2 with v instead of $-v$) to obtain (6.13). We have, for both t'_1 and t'_2 ,

$$t' = \gamma(t - vx/c^2), \quad 0 = x' = \gamma(x - vt).$$

The second of these equations gives $x = vt$, which we knew anyway, and substituting this into the first equation gives

$$\begin{aligned} t' &= \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}} && \text{(by definition of } \gamma) \\ &= \frac{t - v^2t/c^2}{\sqrt{1 - v^2/c^2}} && \text{(using } x = vt) \\ &= t/\gamma \end{aligned}$$

which is the same result as before.



6.4.4 The ladder-and-barn non-paradox

A builder runs towards a barn of length L carrying a ladder of length $2L$ at a speed¹⁹ such that $\gamma = 2$ so that the length contraction factor is $\frac{1}{2}$.

- In the barn's rest frame, the moving ladder undergoes length contraction and has length L . It can therefore fit snugly in the barn.
- In the builder's rest frame, the barn is rushing towards the ladder and undergoes length contraction to $L/2$. There is no way the ladder can fit in.

How can these two statements be reconciled?

The answer stems, as is often the case with apparent paradoxes in relativity, from loose use of language. In this case, it is the use of the word 'fit'; what does it mean to say the ladder 'fits' exactly into the barn? Clearly, we mean that the two events:

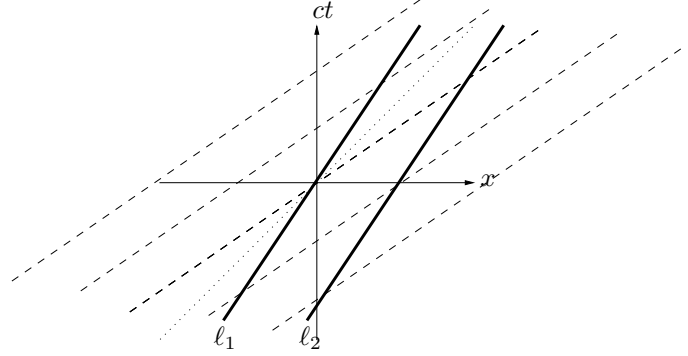
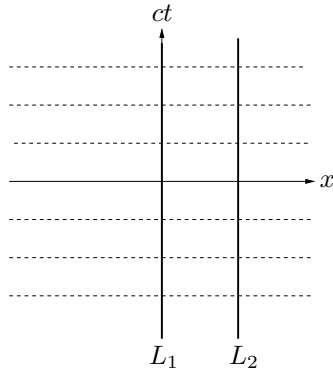
¹⁸I'm not sure that this is a helpful description: what exactly is dilated?? It is better, as always in Special Relativity, to fix on a precise space-time description of the situation: what events we are considering and in which frame.

¹⁹If $v = \sqrt{3}c/2$, then $\gamma^{-2} = 1 - 3/4$ and $\gamma = 2$.

- (i) front end of ladder hits back of barn; (ii) back end of ladder goes through the door

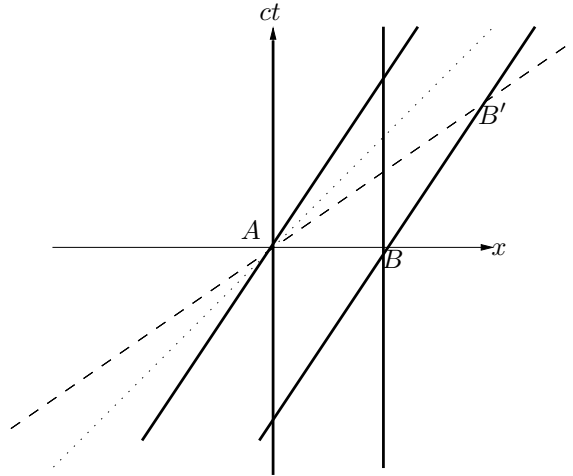
are simultaneous. But observers in different frames do not agree on simultaneity, so ‘fit into’ is a frame-dependent concept: we should not expect observers in different frames to agree so there is no paradox to account for. The two statements are true and compatible and that is really the end of the story. However, we can investigate further.

The situation can best be understood by means of space-time diagrams.



The lines L_1 ($x = 0$) and L_2 ($x = L$) in the left hand figure are the world lines of the ends of the barn in axes corresponding to the rest-frame of the barn; the barn door is L_1 . The lines of simultaneity in the barn frame, $t = \text{constant}$, are shown as broken lines.

The lines ℓ_1 ($x = vt$) and ℓ_2 ($x = vt + L$) in the right hand figure are the world lines of the ends of the ladder, again in the axes corresponding to the rest-frame of the barn. The lines of simultaneity in the ladder frame, $t' = \text{constant}$, i.e. $t - vx/c^2 = \text{constant}$, are shown as broken lines. The light cone is shown as a dotted line.



The previous diagrams are superimposed. The event A is ‘back of ladder goes through barn door’. The event B is ‘front of ladder hits back of barn’. In the barn frame, these events are simultaneous. In the ladder frame, A is simultaneous with B' : by the time A occurs, the front of the ladder has burst through the back of the barn.

Regarded from the point of view of a space-time diagram, the paradox dissolves. One consequence of time not being invariant under Lorentz transformations is that the ladder ‘fits in’ the barn in one frame but does not ‘fit in’ in another.

6.4.5 The twins non-paradox

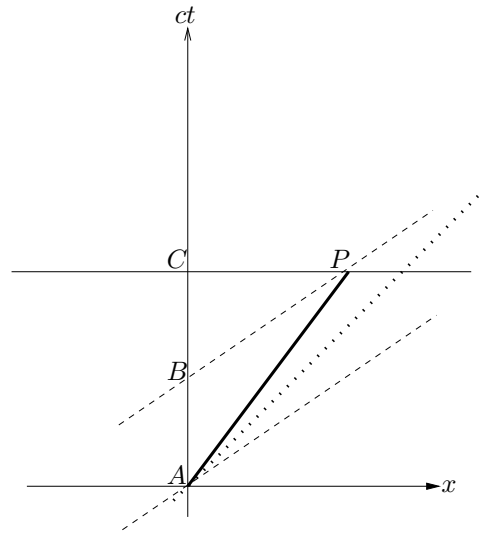
Twins Alice and Bob synchronise watches in an inertial frame and then Bob sets off at speed $\sqrt{3}c/2$, which corresponds to $\gamma = 2$. When Bob has been travelling for a time T according to Alice, he reaches Proxima Centauri²⁰ and turns round by means of accelerations that are very large in his

²⁰The closest star to the Sun: about 4.2 light years away.

frame and goes back to Alice at the same speed. Since Bob is in a moving frame, relative to Alice, his time runs slower by a factor of γ than Alice's, so he will only have aged by $2T \times \frac{1}{2}$ on the two legs of the journey. Thus when they meet up again, Alice has aged by $2T$ but Bob has aged only by T . *This is not the paradox: it is just a fact of life.*²¹

The difficulty some people have with Alice and Bob is the apparent symmetry: surely exactly the same argument could be made, from Bob's point of view, to show that Alice would be the younger when they met again? But the same argument *cannot* be made for Bob because the situation is not symmetric: Alice's frame is inertial, whereas Bob has to accelerate to turn round: while he is accelerating, his frame is not inertial.

BUT, some people might say, suppose we just consider the event of Bob's arrival at Proxima Centauri, so as not to worry about acceleration. Now the situation is symmetric. Surely from Alice's point of view, when Bob arrives he will have aged half as much as Alice, and from Bob's point of view, when he arrives, Alice will have aged half as much as Bob? The answer to this is a simple 'yes'. Surely, they would then say, this doesn't make sense? But it does, as long as you are careful about the word 'when'.



In the above diagram, Alice's world line is the ct (containing points A , B and C) axis and Bob's world line is the line containing A and P . P represents the event 'Bob arrives at Proxima Centauri'.

The line CP is a line of simultaneity in Alice's frame and C is the event 'Alice is at this point in space-time *when* — according to Alice — Bob arrives at Proxima Centauri'; the first use of the word 'when'.

The line BP is a line of simultaneity in Bob's frame and B is the event 'Alice is at this point in space-time *when* — according to Bob — he arrives Proxima Centauri'; the second use of the word 'when'. The two 'whens' don't mean the same thing, since one is a 'when' in Alice's frame the other is a 'when' in Bob's frame.

We can do the calculation. Let us assume for simplicity that Bob sets off the moment he is born. The event C has coordinates $(cT, 0)$ in Alice's frame, and the event P has coordinates (cT, vT) . In Bob's frame, the elapsed time T' is given by the Lorentz transformation:

$$T' = \gamma(T - v^2T/c^2) = T/\gamma = \frac{1}{2}T.$$

This is just the usual time dilation calculation. Thus Bob and Alice agree that Bob's age at Proxima Centauri is $\frac{1}{2}T$. In Alice's frame, Bob has aged half as much as Alice.

We now work out the coordinates of the event B , sticking with Alice's frame. The line of simultaneity, BP has equation $t' = \frac{1}{2}T$, i.e. (using a Lorentz transformation)

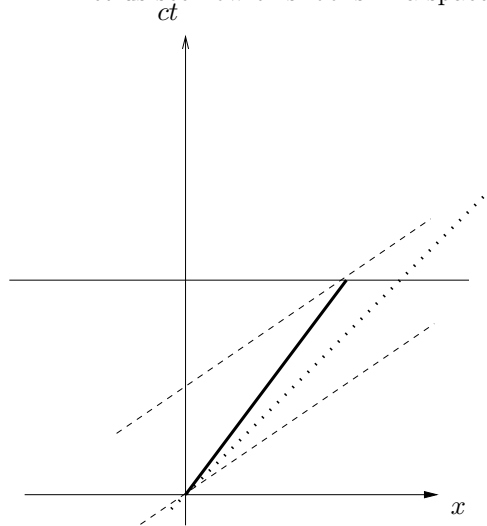
$$\gamma(t + vx/c^2) = \frac{1}{2}T$$

²¹In 1971, Hafele and Keating packed four atomic (caesium) clocks into suitcases and went round the Earth, in different directions, on commercial flights. When they returned, they found that the clocks were slightly behind a clock remaining at the first airport. The result was somewhat inconclusive. The calculations are complicated by the fact that the rate of the clocks is also affected by the gravitational field: clocks run slower in stronger fields, and in fact the two effects balance at $3R/2$ (where R is the radius of the Earth). Thus the heights of the aircraft had to be taken into account as well as their speeds, and it turns out that the two effects are of comparable magnitude, namely of the order of 100 nanoseconds.

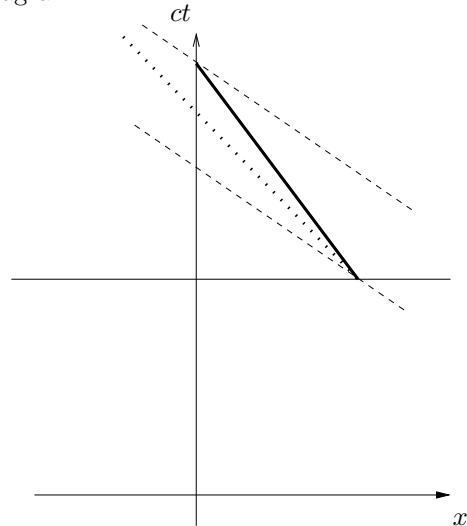
so the point B , for which $x = 0$, has coordinates $(\frac{1}{2}cT/\gamma, 0)$, i.e. $(\frac{1}{4}cT, 0)$. Alice's age when, according to Bob, he arrives at Proxima Centauri is therefore $\frac{1}{4}T$, which is indeed half of Bob's age. So no paradox there either.

BUT, some other people might say, suppose Bob does not turn round but just synchronises his watch at Proxima Centauri with that of another astronaut, Bob', who is going at speed v in the opposite direction (like two trains passing at a station). Each leg of the journey is then symmetric, so why should Alice age faster or slower Bob and Bob' during their legs of the journey? There's no mystery here, either: the situation is indeed symmetric and Alice does indeed age by the same amount as Bob+Bob'. But at the synchronisation event, Bob and Bob' do not agree on Alice's age, because in their different frames the synchronisation event is simultaneous with different times in Alice's life.

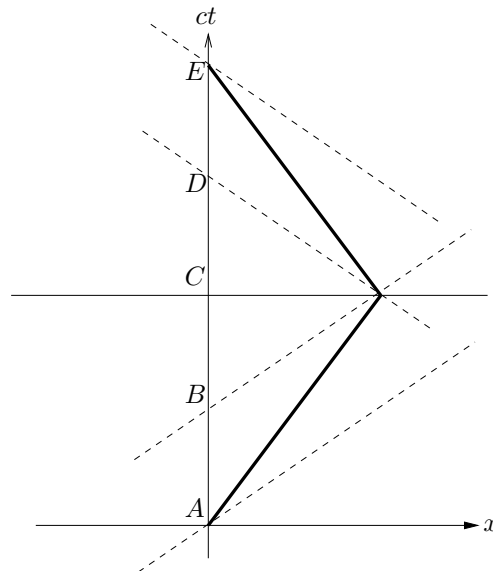
Let us see how this looks in a space-time diagram.



The outward journey. The heavy line is Bob's world line. The dotted line through the origin is the light cone. The dashed lines are the lines of simultaneity in Bob's frame.



The return journey. The heavy line is the world line of Bob'. The dotted line through the turn-round event is the light cone. The dashed lines are the lines of simultaneity in the frame of Bob'.



The superposition of the previous two pictures.

As before, Bob ages by $\frac{1}{2}T$ on the outward journey to Proxima Centauri. By symmetry Bob' ages by $\frac{1}{2}T$ on the inward journey from Proxima Centauri.

However, according to Bob's idea of time, the clock synchronisation occurs when Alice is at B , and according to Bob's it occurs when Alice is at D . Thus Bob's clock will read time T when he meets Alice and Alice's clock will read $2T$. But the time Alice spends between B and D is accounted for by Bob in his journey *after* Proxima Centauri and by Bob' in his journey *before*

reaching Proxima Centauri, so the two Bobs would say that, while they were travelling between Earth and Proxima Centauri, Alice travelled from A to B and then from D to E , taking on her clock a total time T — the same as the journey time of the two Bobs.

Finally, we see that if, instead of meeting Bob', Bob turns round at Proxima Centauri, Alice ages rapidly (according to Bob) from B to D while he is changing direction.

6.5 Velocity transformation

The frame S' moves at constant velocity v with respect to a frame S . A particle P moves with constant velocity u in S . What is the velocity u' of P in S' ?

In Newtonian physics, the answer is simple: $u' = u - v$. However, this cannot be the right answer in Special Relativity because it would imply that the speed of light would not be invariant: $c' = c - v$, where c' is the speed of light in S' .

The world line of P has equation $x = ut$ in S and $x' = u't'$ in S' . We have:

$$\begin{aligned}
 u' &= \frac{x'}{t'} \\
 &= \frac{\gamma(x - vt)}{\gamma(t - vx/c^2)} && \text{(using Lorentz transformations)} \\
 &= \frac{x - vt}{t - vx/c^2} && \text{(cancelling } \gamma) \\
 &= \frac{ut - vt}{t - uv/c^2} && (x = ut, \text{ twice}) \\
 &= \frac{u - v}{1 - uv/c^2} && (6.14)
 \end{aligned}$$

which is the required result. When $uv \ll c^2$, we retrieve the Newtonian result.

If $u = c$ then

$$u' = \frac{c - v}{1 - cv/c^2} = c$$

so the velocity transformation law preserves that speed of light: it is the same to all observers.

Note that this velocity transformation law does not permit a transformation to a speed greater than c . If $u < c$ and $|v| < c$, then

$$\begin{aligned}
 c - u' &= c - \frac{u - v}{1 - uv/c^2} \\
 &= \frac{c(1 - uv/c^2) - (u - v)}{1 - uv/c^2} \\
 &= \frac{c(c - u)(c + v)}{c^2 - uv} > 0.
 \end{aligned}$$

Thus $u' < c$. Similarly, we can show that $-u' < c$ if $-u < c$, so $|u'| < c$ if $|u| < c$ (assuming $|v| < c$).

If we write the velocities u , u' and v in terms of hyperbolic angles:

$$\frac{u}{c} = \tanh \beta, \quad \frac{u'}{c} = \tanh \beta', \quad \frac{v}{c} = \tanh \alpha$$

and substitute into the velocity transformation law (6.14) we find

$$\tanh \beta' = \frac{\tanh \beta - \tanh \alpha}{1 - \tanh \alpha \tanh \beta} = \tanh(\beta - \alpha)$$

so an alternative form of the transformation law is

$$\beta' = \beta - \alpha$$

i.e.

$$\tanh^{-1}(u'/c) = \tanh^{-1}(u/c) - \tanh^{-1}(v/c).$$

6.6 Proper time

6.6.1 Definition

The fact that the concept of time is frame dependent can be rather unsettling. It would be good to have some quantity that corresponds to time but does not vary at the whim of the observer. Such a quantity exists and is called *proper time*.

We define the proper time $\Delta\tau$ between two events E_1 and E_2 on the world line of an observer, with coordinates (ct, x) and $(ct + c\Delta t, x + \Delta x)$ respectively, by

$$c^2(\Delta\tau)^2 = c^2(\Delta t)^2 - (\Delta x)^2. \quad (6.15)$$

For this to make sense, we require either $c\Delta t \geq |\Delta x|$ or $c\Delta t \leq -|\Delta x|$, which means that either E_1 is in the past light cone of E_2 , or vice versa (which is why we specified that the events are on the world line of an observer). In the former case, we choose $\Delta\tau \geq 0$.

Note that $(\Delta\tau)^2 > 0$ if the vector joining E_1 and E_2 is time-like, and $(\Delta\tau)^2 = 0$ if the vector joining E_1 and E_2 is null. For points joined by a space-like vector, one can define *proper distance*, s , by

$$(\Delta s)^2 = (\Delta x)^2 - c^2(\Delta t)^2.$$

In the rest frame of the observer, $\Delta x = 0$, and $\Delta\tau = \Delta t$. Thus proper time measures rest frame time.

We can verify that proper time is Lorentz invariant by brute force. Since Lorentz transformations are linear, we have

$$\begin{aligned} \Delta t' &= \gamma(\Delta t - v\Delta x/c^2) \\ \Delta x' &= \gamma(\Delta x - v\Delta t). \end{aligned} \quad (6.16)$$

Therefore

$$\begin{aligned} c^2(\Delta\tau')^2 &\equiv c^2(\Delta t')^2 - (\Delta x')^2 \\ &= c^2(\gamma(\Delta t - v\Delta x/c^2))^2 - (\gamma(\Delta x - v\Delta t))^2 \\ &= \gamma^2(c^2(\Delta t)^2 - v^2(\Delta t)^2) + \gamma^2(v^2(\Delta x)^2/c^2 - (\Delta x)^2) && \text{(the cross terms cancel)} \\ &= c^2(\Delta t)^2 - (\Delta x)^2 && (\gamma^2 = (1 - v^2/c^2)^{-1}) \\ &= c^2(\Delta\tau)^2 && \text{(as required)} \end{aligned}$$

However, we can most easily verify that this quantity is Lorentz invariant using the matrix form of the Lorentz transformation:

$$\begin{aligned} c^2(\Delta\tau)^2 &\equiv (c\Delta t, \Delta x) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} c\Delta t \\ \Delta x \end{pmatrix} && (6.17) \\ &= (c\Delta t, \Delta x) \eta \begin{pmatrix} c\Delta t \\ \Delta x \end{pmatrix} && \text{(by definition (6.11))} \\ &= (c\Delta t, \Delta x) L^T \eta L \begin{pmatrix} c\Delta t \\ \Delta x \end{pmatrix} && \text{(see (6.12))} \\ &= (c\Delta t', \Delta x') \eta \begin{pmatrix} c\Delta t' \\ \Delta x' \end{pmatrix} \\ &= c^2(\Delta t')^2 - (\Delta x')^2 \\ &\equiv c^2(\Delta\tau')^2. \end{aligned}$$

The proper time between two infinitesimally separated points (ct, x) and $(ct + cdt, x + dx)$ is given by

$$c^2 d\tau^2 = c^2 dt^2 - dx^2. \quad (6.18)$$

Again, we note that if these points represent events on the world line of an observer then in the rest frame of the observer

$$d\tau = dt_{\text{rest}}$$

so infinitesimal proper time measures infinitesimal time displacements in the rest frame; ticks of the observer's clock. Comparing with (6.18) we see that in a general frame

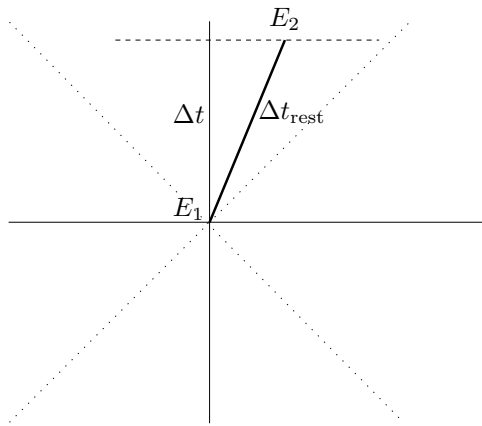
$$\sqrt{dt^2 - dx^2/c^2} = d\tau = dt_{\text{rest}}$$

so

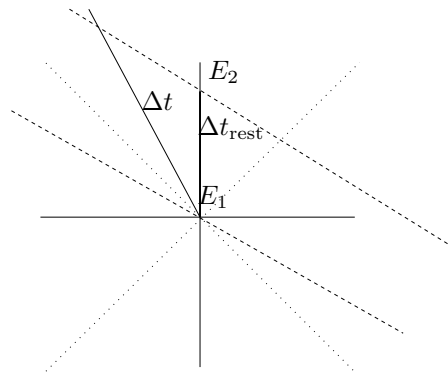
$$dt > dt_{\text{rest}}$$

which is time dilation.

Rather confusingly, this comparison cannot be easily inferred from a space-time diagram: in fact, there is a temptation (which must be resisted) to assume that lengths behave in a Euclidean fashion (so that Pythagoras applies) which would lead to completely incorrect conclusions²². This is illustrated in the following diagrams. In both diagrams, the null cone is represented by the pair of dotted lines.



E_1 and E_2 are events on the world line of an observer represented in this diagram as moving with respect to a 'general' observer. The horizontal dashed line is simultaneity for the 'general' observer. $\Delta t_{\text{rest}} < \Delta t$, even though it doesn't look as if this is the case.



E_1 and E_2 are events on the world line of an observer represented in this diagram as stationary. The world line of the 'general' observer is slanted. The dashed lines are simultaneity for the 'general' observer. $\Delta t_{\text{rest}} < \Delta t$, which does look to be the case, though the difference in times is not as much as it 'seems'.

Now suppose that these two points represent events on the world line of an observer moving with velocity v with respect to some given frame (the 'lab frame'). The world line can be written in the form

$$x = x(t)$$

in which case

$$v = \frac{dx}{dt}.$$

Then

$$\begin{aligned} d\tau^2 &= dt^2 \left(1 - \frac{1}{c^2} \frac{(dx)^2}{(dt)^2} \right) \\ &= dt^2 \left(1 - \frac{v^2}{c^2} \right) \end{aligned}$$

and

$$\frac{dt}{d\tau} = \gamma. \quad (6.19)$$

Note that here γ is not linked to the velocity between two frames explicitly, though of course it is implicitly related to the velocity between the rest frame of the observer and the lab frame.

²²You wouldn't dream of using Pythagoras in an A-level type distance-time graph; nevertheless, it is a good instinct to suppose that there is some concept of length that can be applied to a space-time diagram in Special Relativity and this is touched on in the next section.

The total time that elapses on the world-line of an observer moving with (not necessarily constant) velocity in a frame S is given by

$$\int d\tau = \int \gamma^{-1} dt;$$

this is the observer's actual time (clock or biological).

We can use proper time to derive the velocity addition formula (6.14) for an observer moving with non-constant velocity. We parameterise the observer's world line by τ :

$$\begin{aligned} x &= x(\tau), \quad t = t(\tau) && \text{in } S \\ x' &= x'(\tau), \quad t' = t'(\tau) && \text{in } S' \end{aligned}$$

and

$$u = \frac{dx}{d\tau} \bigg/ \frac{dt}{d\tau}, \quad u' = \frac{dx'}{d\tau} \bigg/ \frac{dt'}{d\tau},$$

We can differentiate the Lorentz transformation (6.2) and (6.1) to obtain

$$\begin{aligned} \frac{dx'}{d\tau} &= \gamma \left(\frac{dx}{d\tau} - v \frac{dt}{d\tau} \right) = \gamma(u - v) \frac{dt}{d\tau} \\ \frac{dt'}{d\tau} &= \gamma \left(\frac{dt}{d\tau} - (v/c^2) \frac{dx}{d\tau} \right) = \gamma(1 - (uv/c^2)) \frac{dt}{d\tau} \end{aligned} \quad (6.20)$$

and dividing these expressions gives

$$u' = \frac{u - v}{1 - uv/c^2}$$

as required. ²³

6.6.2 Line elements and metrics

The infinitesimal version of the formula (6.17) for proper time

$$c^2 d\tau^2 \equiv (cdt, \quad dx) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} cdt \\ dx \end{pmatrix}$$

can be written in suffix notation

$$c^2 d\tau^2 = \eta_{ij} dx^i dx^j, \quad (6.21)$$

where the infinitesimal vector²⁴ dx^i has components (cdt, dx) . The matrix η in the expression (6.21) is called the *metric*: it tells us how the invariant distance can be calculated from the coordinates. The whole of the right hand side of equation (6.21) is called the *line element*.

In two dimensional Euclidean space in Cartesian coordinate,

$$ds^2 = dx^2 + dy^2,$$

which corresponds to a metric which is the unit 2×2 matrix. In plane polar coordinates,

$$ds^2 = dr^2 + r^2 d\theta^2$$

so the metric is

$$\begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}.$$

On the surface of a sphere of radius a , infinitesimal distances are given by

$$ds^2 = a^2 d\theta^2 + a^2 \sin^2 \theta d\phi^2,$$

corresponding to the metric

$$\begin{pmatrix} a^2 & 0 \\ 0 & a^2 \sin^2 \theta \end{pmatrix}.$$

This metric describes a curved surface, and the curvature and other properties of the surface can be deduced from the metric.

²³Note to self: how about adding something about the action of the Lorentz group on space-time (the orbits are hyperbolas given by $(ct)^2 - x^2 = \text{constant}$, etc).

²⁴The index is conventionally placed upstairs (superscript), rather than downstairs (subscript) as in normal vector calculus. The reason for this is related to the way that this vector transforms under Lorentz transformations. We will not need to worry about this in this course, but it seems best to use the correct convention.

6.7 Constant acceleration in Special Relativity

This section is not examinable and is included for interest

It is often said, erroneously, that Special Relativity cannot deal with acceleration because it deals only with inertial frames, and that therefore acceleration must be the preserve of General Relativity. We must, of course, only allow transformations between inertial frames; the frames must not accelerate, but the observers in the frame can move as the please. Special Relativity can deal with anything kinematic but General Relativity is required when gravitational forces are present.

As an example of non-uniform motion, we consider an observer who is moving with constant acceleration.

The first step is to define what we mean by ‘constant acceleration’ which is certainly a frame-dependent concept. The most common situation is that of an observer in a rocket experiencing a constant ‘G-force’ due to the rocket thrust. This corresponds to the acceleration measured in the instantaneous (inertial) rest frame of the rocket being constant (acceleration having the usual definition of dv/dt), so we take this to be our definition.

For reasons that will later become clear, we need to determine the way that acceleration transforms under Lorentz transformations. We can do this in a number of ways. We will here start with the velocity transformation law (6.20) for an observer with world line given in S by $(ct(\tau), x(\tau))$ and in S' by $(ct'(\tau), x'(\tau))$. Forgetting the acceleration problem for the moment, we assume that these frames have a constant relative velocity v .

The velocities u and u' in the two frames are related by

$$u' = \frac{u - v}{1 - uv/c^2} \equiv \frac{(c^2/v)(1 - v^2/c^2)}{1 - uv/c^2} - \frac{c^2}{v}$$

(the equivalent form is just a bit of algebra to obtain a useful expression). Differentiating this with respect to τ gives

$$\frac{du'}{d\tau} = \frac{1 - v^2/c^2}{(1 - uv/c^2)^2} \frac{du}{d\tau}. \quad (6.22)$$

The acceleration, a , in S is by definition du/dt and similarly for S' so

$$\begin{aligned} a' &= \frac{du'}{dt'} \\ &= \frac{du'}{d\tau} \bigg/ \frac{dt'}{d\tau} \\ &= \frac{1 - v^2/c^2}{(1 - uv/c^2)^2} \frac{du}{d\tau} \bigg/ \frac{dt'}{d\tau} && \text{(using (6.22))} \\ &= \frac{1 - v^2/c^2}{(1 - uv/c^2)^2} \frac{du}{d\tau} \bigg/ \gamma(1 - uv/c^2) \frac{dt}{d\tau} && \text{(using (6.20))} \\ &= \frac{(1 - v^2/c^2)^{\frac{3}{2}}}{(1 - uv/c^2)^3} a. \end{aligned} \quad (6.23)$$

As mentioned above there are other ways of obtaining this result; for example, more elegantly using four-vectors (see section 6.7).

In the situation we have in mind, S' is the instantaneous rest frame of the accelerating observer, so that $u' = 0$ and $u = v$, and the acceleration a' in this frame is constant (i.e. independent of v). Thus (6.23) becomes

$$a = (1 - u^2/c^2)^{\frac{3}{2}} a'$$

Now

$$a = \frac{du}{d\tau} \bigg/ \frac{dt}{d\tau} \quad \text{and} \quad \frac{dt}{d\tau} = (1 - u^2/c^2)^{-\frac{1}{2}}$$

so we can find the parameterised equation of the world line by integrating

$$\frac{du}{d\tau} = a \frac{dt}{d\tau} = (1 - u^2/c^2) a'.$$

This gives

$$u = c \tanh(a'\tau/c) \quad \text{(choosing the origin of } \tau \text{ so that } u = 0 \text{ when } \tau = 0)$$

and hence

$$\gamma = \cosh(a'\tau/c).$$

Then from $dt/d\tau = \gamma$, we find that

$$t = c/a' \sinh(a'\tau/c) \quad (\text{choosing the origin of } t \text{ such that } t = 0 \text{ when } \tau = 0)$$

Finally,

$$\begin{aligned} \frac{dx}{d\tau} &= \frac{dx}{dt} \frac{dt}{d\tau} \\ &= u\gamma \\ &= c \sinh(a'\tau/c) \end{aligned}$$

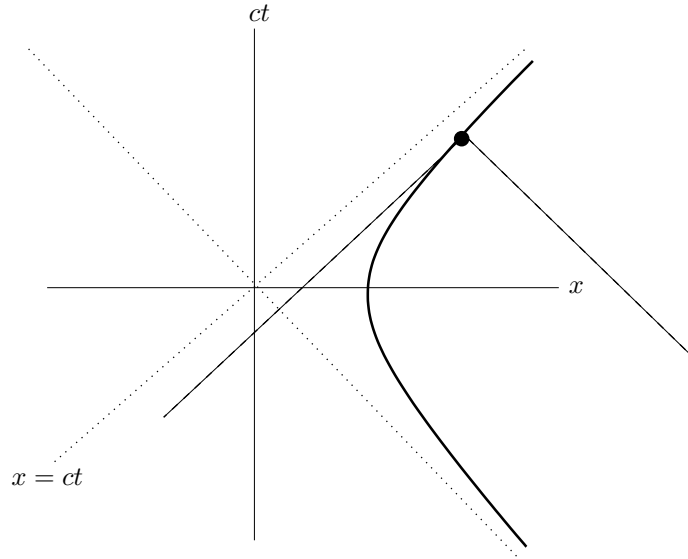
so

$$x = c^2/a' \cosh(a'\tau/c). \quad (\text{choosing the origin of } x \text{ such that } x = c^2/a' \text{ when } t = 0)$$

Uniformly accelerated particles therefore move on rectangular hyperbolas of the form

$$x^2 - (ct)^2 = (c^2/a')^2.$$

The diagram shows the trajectory. The dotted lines are the light cones. An event taking place within the dashed lines can influence an accelerated observer at the position shown, but events taking place outside the dashed lines would have to move faster than the speed of light to do so. As $\tau \rightarrow \infty$, the whole of the space-time to the left of the dotted line $x = ct$ would be inaccessible to the observer. This line is called the *Rindler event horizon* for the accelerated observer. In some ways, it performs the same function as the event horizon of a black hole. In particular, the observer has to accelerate to avoid falling through it and anything happening on the other side would be hidden to the observer. Of course, the accelerating observer could just stop accelerating whereas the observer in a black hole space-time can do nothing to affect the event horizon.



The space-time diagram for an accelerated observer. The thick hyperbola is the observer's world line. An observer 'below' the dashed lines could in principle send a message to the observer marked as a heavy dot; other observers could not.

6.8 Four-vectors

6.8.1 definitions

In 1 + 3 dimensions (i.e. one time dimension and three space dimensions), we write the position 4-vector X in the form

$$X = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

or sometimes

$$X = \begin{pmatrix} ct \\ \mathbf{x} \end{pmatrix}.$$

A (4×4) matrix L representing a transformation from a frame S to a frame S'

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} \equiv X' = LX \quad (6.24)$$

is a Lorentz transformation if

$$L^T \eta L = \eta \quad (6.25)$$

where η is the diagonal matrix (the Minkowski metric) defined by

$$\eta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

This is exactly as in the $(1+1)$ dimensional case (6.11).

Lorentz transformations determine a mapping from an inertial frame S to a new frame S' via the mapping (6.24) regarded as a coordinate transformation. We can easily verify that S' is inertial and that the speed of light is preserved.²⁵

The Lorentz transformations form a matrix group, as can easily be verified from the definition (6.25). It is called $O(1, 3)$. It includes reflections (including $t \rightarrow -t$), which one usually wants to exclude. The subgroup of the Lorentz group consisting of matrices that satisfy $\det L = +1$ is called the *proper Lorentz group* and denoted $SO(1, 3)$. Those which preserve the orientation of the spatial axes and the direction of time form the *restricted Lorentz group*²⁶, denoted $SO^+(1, 3)$.

It is at first sight rather difficult to understand what sort of matrices might satisfy the definition (6.25) of a Lorentz transformation matrix. However, we can make some progress by counting arguments. A 4×4 matrix has 16 potentially independent elements. In this case they are constrained by the 10 independent equations of (6.25): there are 10 equations, not 16, because both the left hand side and the right hand side are symmetric matrices. Thus just 6 of the 16 elements of L are independent. We express this by saying that the Lorentz matrices form a 6-parameter group.

This group includes rotations of the spatial axes:

$$L = \left(\begin{array}{c|ccc} 1 & 0 & 0 & 0 \\ \hline 0 & & & \\ 0 & R & & \\ 0 & & & \end{array} \right)$$

where R is 3×3 rotation matrix (so that $R^T I R = I$). These rotations form a 3-parameter subgroup, each parameter being (for example) an angle of rotation about one of the three spatial axes.

The group also includes ‘boosts’ of the form

$$L = \left(\begin{array}{cc|cc} \gamma & -v\gamma/c & 0 & 0 \\ -\gamma v/c & \gamma & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right). \quad (6.26)$$

This represents a boost in the x direction, but it could equally well have been a boost in the y or z directions. Each of these boosts forms a 1-parameter subgroup (the parameter being the velocity along the relevant axis), so that is another three parameters.

Thus the boosts and rotations together require 6 independent matrix elements, which is the total number available. This indicates that any Lorentz matrix is a combination of rotations and boosts.

We will not be interested in rotations. In fact, we will restrict our attention to transformations of the form (6.26). We do not lose much generality by this restriction, because any proper Lorentz

²⁵I should do this explicitly later on.

²⁶The restricted Lorentz group is isomorphic to the Möbius group, and this fact is the basis of the twistor theory devised by Roger Penrose as a possible route to quantum gravity.

transformation can be expressed as a spatial rotation followed by a transformation of the form (6.26); this means that we could choose axes (by a spatial rotation) such that the required Lorentz transformation is exactly (6.26).

A *scalar* or *invariant* under Lorentz transformations is a quantity that is the same in any two frames related by a Lorentz transformation. One example of a Lorentz scalar is the proper time between two events in space-time; another is the rest mass of a particle, which is the mass measured in the particle's rest frame and therefore is invariant by definition.

A *4-vector* V is a quantity whose components transform according to the rule

$$\begin{pmatrix} V'_0 \\ V'_1 \\ V'_2 \\ V'_3 \end{pmatrix} = L \begin{pmatrix} V_0 \\ V_1 \\ V_2 \\ V_3 \end{pmatrix}$$

where L is the matrix of the Lorentz transformation. The position 4-vector transforms as a 4-vector by definition of the transformation (6.24) itself.

Given any two 4-vectors X and Y , we define an inner product called the *scalar product* by

$$X \cdot Y = X^T \eta Y \equiv X_0 Y_0 - X_1 Y_1 - X_2 Y_2 - X_3 Y_3$$

or if

$$X = \begin{pmatrix} X_0 \\ \mathbf{x} \end{pmatrix} \quad \text{and} \quad Y = \begin{pmatrix} Y_0 \\ \mathbf{y} \end{pmatrix}$$

then

$$X \cdot Y = X_0 Y_0 - \mathbf{x} \cdot \mathbf{y}.$$

The scalar product between two 4-vectors is Lorentz invariant:

$$\begin{aligned} X' \cdot Y' &= (X')^T \eta Y' && \text{(by definition of the scalar product)} \\ &= (LX)^T \eta LY && \text{(transforming back to the unprimed frame)} \\ &= X^T L^T \eta LY \\ &= X^T \eta Y && \text{(because } L \text{ is a Lorentz transformation)} \\ &= X \cdot Y && \text{(by definition of the scalar product)} \end{aligned}$$

The scalar product of a 4-vector with itself (the 'length squared') can be negative, positive or zero. Since the scalar product is invariant, this classifies the vectors into three classes:

$$\begin{aligned} V \cdot V &> 0 && V \text{ is time-like} \\ V \cdot V &< 0 && V \text{ is space-like} \\ V \cdot V &= 0 && V \text{ is null} \end{aligned}$$

In particular, the quantity $dX \cdot dX$ is invariant, where dX is the infinitesimal 4-vector connecting two neighbouring space-time points. If dX is time-like or null, we define the infinitesimal proper time, $d\tau$, up to a sign, by

$$c^2 d\tau^2 = dX \cdot dX = c^2 dt^2 - dx^2 - dy^2 - dz^2 = c^2 dt^2 - d\mathbf{x} \cdot d\mathbf{x}. \quad (6.27)$$

In the case of a time-like vector, we choose $d\tau > 0$ if $dt > 0$.

6.8.2 4-velocity

Now consider the infinitesimal position vector dX joining two points on the world line $X = X(\tau)$ of an observer. Since $d\tau$ is invariant and dX is a 4-vector, we can form another 4-vector, the *4-velocity* U of the observer, by dividing them:

$$U = \frac{dX}{d\tau} = \begin{pmatrix} c \frac{dt}{d\tau} \\ \frac{d\mathbf{x}}{d\tau} \end{pmatrix}. \quad (6.28)$$

In the rest frame of the observer, $\mathbf{x} = \mathbf{0}$ and $d\tau = dt$ so

$$U = \begin{pmatrix} c \\ \mathbf{0} \end{pmatrix} \quad (6.29)$$

again in the rest frame of the observer.

Now we have the first demonstration of the power of the 4-vector formalism. The length squared of the velocity 4-vector is

$$\begin{aligned}
 U \cdot U &= c^2 \left(\frac{dt}{d\tau} \right)^2 - \frac{d\mathbf{x}}{d\tau} \cdot \frac{d\mathbf{x}}{d\tau} \\
 &= c^2 \left(\frac{dt}{d\tau} \right)^2 - \frac{d\mathbf{x}}{dt} \cdot \frac{d\mathbf{x}}{dt} \left(\frac{dt}{d\tau} \right)^2 \\
 &= c^2 \left(\frac{dt}{d\tau} \right)^2 \left(1 - \frac{\mathbf{u} \cdot \mathbf{u}}{c^2} \right) \\
 &= c^2 \left(\frac{dt}{d\tau} \right)^2 \frac{1}{\gamma^2}
 \end{aligned} \tag{6.30}$$

But in the rest frame,

$$U \cdot U = c^2 \quad (\text{from (6.29)})$$

and since the scalar product is the same in all frames, we can equate this with (6.30):

$$c^2 \left(\frac{dt}{d\tau} \right)^2 = c^2$$

i.e.

$$\frac{dt}{d\tau} = \gamma \tag{6.31}$$

(recall that we choose $d\tau$ to have the same sign as dt .)²⁷

We can use this result to tidy up the expression (6.28) for the 4-velocity:

$$\begin{aligned}
 U &= \begin{pmatrix} c \frac{dt}{d\tau} \\ \frac{d\mathbf{x}}{d\tau} \end{pmatrix} \\
 &= \begin{pmatrix} c \frac{dt}{d\tau} \\ \frac{d\mathbf{x}}{dt} \frac{dt}{d\tau} \end{pmatrix} \\
 &= \begin{pmatrix} c\gamma \\ \mathbf{u}\gamma \end{pmatrix} \quad (\text{using } \frac{dt}{d\tau} = \gamma) \\
 &= \gamma \begin{pmatrix} c \\ \mathbf{u} \end{pmatrix}.
 \end{aligned} \tag{6.32}$$

6.8.3 4-momentum

The rest mass of a particle, defined in the obvious way as the mass of the particle in its rest frame, is Lorentz invariant by its definition.²⁸ For a particle of rest mass m and 4-velocity U , we define the momentum 4-vector, or the *4-momentum*, P by

$$P = mU = \begin{pmatrix} mc\gamma \\ m\gamma\mathbf{u} \end{pmatrix} = \begin{pmatrix} E/c \\ \mathbf{p} \end{pmatrix}$$

where, by definition the relativistic energy E and the relativistic 3-momentum $\mathbf{p}$ are

$$E = m\gamma c^2 \quad \text{and} \quad \mathbf{p} = m\gamma\mathbf{u}$$

²⁷Actually, we could have obtained the same result in a line from the definition of (6.27) proper time by dividing by dt^2 but that shouldn't prevent us from admiring the neatness of the above calculation.

²⁸Note to self: probably better to define relativistic three-momentum and energy earlier? Together with proper time?

Note that the mass here is the rest mass: we will not use any other concept for mass.²⁹

The justification for calling the quantity $m\gamma c^2$ the relativistic energy is three-fold. First, we will see later that it is conserved in a wide variety of circumstances, as one would hope. Second, in the Newtonian limit $v/c \ll 1$, it approximates the kinetic energy plus a residual energy:

$$\begin{aligned} E &= mc^2(1 - v^2/c^2)^{-\frac{1}{2}} \\ &\approx mc^2(1 + \tfrac{1}{2}v^2/c^2) && \text{(using the binomial expansion)} \\ &= mc^2 + \tfrac{1}{2}mv^2. \end{aligned}$$

Finally, we can justify defining relativistic energy in this way by considering its rate of change in terms of work done by forces; this will be done in the next section.

Similarly, the relativistic 3-momentum approximates the non-relativistic momentum in the Newtonian limit (setting $\gamma \approx 1$).

We can again perform the trick of comparing the length squared of this 4-vector in the general frame and in the rest frame. In the rest frame,

$$P = \begin{pmatrix} mc \\ \mathbf{0} \end{pmatrix}$$

so

$$P \cdot P = m^2 c^2.$$

In the general frame

$$P \cdot P = \frac{E^2}{c^2} - \mathbf{p} \cdot \mathbf{p}.$$

Equating these two expressions gives the relationship between relativistic energy and relativistic 3-momentum:

$$E^2 = p^2 c^2 + m^2 c^4 \tag{6.33}$$

where $p^2 = \mathbf{p} \cdot \mathbf{p}$. We could of course have worked this out more laboriously by eliminating u from

$$E = m\gamma c^2, \quad p^2 = m^2 \gamma^2 u^2, \quad \gamma^{-2} = 1 - u^2/c^2.$$

6.8.4 Relativistic energy

We seek to justify defining the relativistic energy by $E = m\gamma c^2$, where m is the rest mass, γ is the usual factor associated with the speed v of the particle, $\gamma = (1 - v^2/c^2)^{-\frac{1}{2}}$.

First we consider the Newtonian situation. For a particle of mass m moving in a force field $\mathbf{F}$ in Newtonian dynamics, we have

$$\text{work done} = \text{force} \times \text{distance moved by force} = \int \mathbf{F} \cdot d\mathbf{x} = \int \mathbf{F} \cdot \mathbf{v} dt$$

and hence (using Newton's second law)

$$\text{rate of doing work} = \mathbf{F} \cdot \mathbf{v} = \frac{d\mathbf{p}}{dt} \cdot \mathbf{v} = \frac{d}{dt} \left(\tfrac{1}{2} m \mathbf{v} \cdot \mathbf{v} \right) = \frac{dT}{dt},$$

where T is the kinetic energy of the particle. Thus the rate of doing work on the particle is equal to the increase in its kinetic energy.

We now look at the relativistic situation. In a general inertial frame, the 4-momentum P is given by

$$P = (E/c, \mathbf{p})$$

²⁹Sometimes, the concept of relativistic mass is introduced. If m_0 is the rest mass of a particle moving with velocity $\mathbf{v}$, then its relativistic mass is $m_0\gamma$. This leads to the famous energy-mass equivalence equation $E = mc^2$. It is a shame not to be able to use this equation, but it does seem terribly profligate to have two names (E and mc^2) for essentially the same quantity: relativistic energy. It avoids confusion to have just one sort of mass, namely rest mass. This was also Einstein's view, stated many times, despite his name being eternally linked to the famous equation. In his 'Annus mirabilis' (1905) when he wrote 5 papers each of which would have made his name (size of molecules, when it wasn't widely accepted that molecules existed; Brownian motion, evidence for the existence of molecules; viscosity formula; on the motion of electromagnetic bodies, introducing Special Relativity; inertia of a body) he related the change of mass to change of energy, but did not explicitly give the energy of the body as mc^2 .

and $P \cdot P = m^2 c^2$ which is constant. Thus

$$\begin{aligned}
 0 &= \frac{d}{dt} (P \cdot P) && \text{(because } P \cdot P = m^2 c^2 \text{)} \\
 &= \frac{d}{dt} (E^2/c^2 - \mathbf{p} \cdot \mathbf{p}) \\
 &= 2 \frac{E}{c^2} \frac{dE}{dt} - 2 \mathbf{p} \cdot \frac{d\mathbf{p}}{dt} \\
 &= 2m\gamma \frac{dE}{dt} - 2m\gamma \mathbf{v} \cdot \frac{d\mathbf{p}}{dt} && \text{(since } E = m\gamma c^2 \text{ and } \mathbf{p} = m\gamma \mathbf{v} \text{)} \\
 &= 2m\gamma \left(\frac{dE}{dt} - \mathbf{v} \cdot \frac{d\mathbf{p}}{dt} \right)
 \end{aligned}$$

so

$$\frac{dE}{dt} = \mathbf{v} \cdot \mathbf{F} \quad (6.34)$$

where $\mathbf{F}$ is the force acting on the particle and causing its momentum to change. The last inequality follows from one of the basic postulates of Special Relativity, namely that Newton's law holds in inertial frames provided the momentum is correctly defined. Equation (6.34) shows that the rate of change of relativistic energy is equal to the rate of doing work, which helps to justify the definition $E = m\gamma c^2$.

6.8.5 Massless particles

The (rest) mass of a particle in Special Relativity is defined invariantly by the length of its 4-momentum vector

$$P \cdot P = m^2 c^2$$

or, equivalently, in terms of its relativistic energy and 3-momentum by $E^2 - (\mathbf{p} \cdot \mathbf{p})c^2 = m^2 c^4$. Thus a particle can be massless if and only if its 4-momentum is null (i.e. light-like); the momentum 4-vector points along the instantaneous null cone of the particle.³⁰ In that case,

$$P \cdot P = 0, \quad \text{i.e.} \quad P_0^2 = P_1^2 + P_2^2 + P_3^2 = \mathbf{p} \cdot \mathbf{p}.$$

We can therefore write the 4-momentum in the form

$$P = p(1, \mathbf{k}) \equiv (E/c)(1, \mathbf{k})$$

where $\mathbf{k}$ is unit vector and $p^2 = \mathbf{p} \cdot \mathbf{p}$.

The only massless particles we consider here are photons. There are others (or may be).³¹ If we use the relation from quantum mechanics between frequency and energy of a photon

$$E = h\nu \equiv hc/\lambda$$

where ν is frequency and λ is wavelength, we can write the 4-momentum as

$$P = h\nu(1, \mathbf{k}).$$

Photons (and other massless particles) move with the speed of light, so that for the world line of a photon,³²

$$d\tau^2 = c^2 dt^2 - d\mathbf{x} \cdot d\mathbf{x} = 0.$$

This means that we cannot construct a 4-velocity vector for a massless particle by means of

$$V = \frac{dX}{d\tau}.$$

³⁰Note to self: perhaps expand this section to explain the wave mechanics and justify that P in this case is a 4-vector.

³¹Gluons, which form quarks, are massless. They are predicted by quantum chromodynamics and have been indirectly detected but not observed. Neutrinos were thought to be massless, but are now thought to have mass; a very small mass (luckily, because 50 trillion solar electron neutrinos pass through the human body every second).

³²The fact that proper time is zero along the world line of a massless particle is taken on Physics Forum and similar web sites to imply a negative answer the question 'do photons experience time?'; but this question seems to me to be akin to the question 'do slugs feel pain?'.

6.8.6 Transformations of 4-vectors

Choosing axes such that the velocity of the inertial frame $\mathcal{S}'$ with respect to the inertial frame $\mathcal{S}$ is along the x axis, the Lorentz transformation matrix can be written in the form (6.26)

$$\begin{pmatrix} \gamma & -\gamma v/c & 0 & 0 \\ -\gamma v/c & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (6.35)$$

where $\gamma = (1 - v^2/c^2)^{-\frac{1}{2}}$.

Let U be the 4-velocity of a particle P . We will calculate the effect of a transformation from a frame $\mathcal{S}$ to a frame $\mathcal{S}'$ when the 3-velocity $\mathbf{v}$ of $\mathcal{S}'$ relative to $\mathcal{S}$ is not aligned with the 3-velocity $\mathbf{u}$ of P . In $\mathcal{S}$ we can write U without loss of generality (i.e. by making a rotation of axes, if necessary, in the y - z plane, so that the z axis is orthogonal to $\mathbf{u}$) as

$$\begin{pmatrix} \gamma_u c \\ \gamma_u u \cos \alpha \\ \gamma_u u \sin \alpha \\ 0 \end{pmatrix}. \quad (6.36)$$

where $u = |\mathbf{u}|$, and we have defined the gamma factor associated with the 4-velocity U in the normal way:

$$\gamma_u = \frac{1}{(1 - u^2/c^2)^{\frac{1}{2}}}.$$

Expressing the components of U in $\mathcal{S}'$ in the same form as (6.36) and using the Lorentz transform (6.35) gives

$$\begin{pmatrix} \gamma_{u'} c \\ \gamma_{u'} u' \cos \alpha' \\ \gamma_{u'} u' \sin \alpha' \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma v/c & 0 & 0 \\ -\gamma v/c & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma_u c \\ \gamma_u u \cos \alpha \\ \gamma_u u \sin \alpha \\ 0 \end{pmatrix}, \quad (6.37)$$

from which formulae that determine u' and α' can be read off. Writing out the three components gives

$$\gamma_{u'} = \gamma \gamma_u (1 - (uv/c^2) \cos \alpha) \quad (6.38)$$

$$\gamma_{u'} u' \cos \alpha' = \gamma \gamma_u (-v + u \cos \alpha) \quad (6.39)$$

$$\gamma_{u'} u' \sin \alpha' = \gamma_u u \sin \alpha \quad (6.40)$$

It is all a bit congested, but if we divide (6.39) by (6.38) something nice happens:

$$u' \cos \alpha' = \frac{u \cos \alpha - v}{1 - uv \cos \alpha / c^2}.$$

This shows that the x component of the velocity (which is the component parallel to $\mathbf{v}$) transforms exactly as in the 2-dimensional case (6.14). We could of course obtain this result in identical form to (6.14) by simply setting $\cos \alpha = 1$ in the matrix equation (6.37).

The second component gives

$$\gamma_{u'} u' = \gamma \gamma_u (-v + u)$$

which doesn't quite look the same as (6.14). However, the two can be reconciled with a few lines of algebra³³. Alternatively, if we look at the first component of (6.37) we find a useful identity for $\gamma_{u'}$:

$$\gamma_{u'} = \gamma \gamma_u (1 - uv/c^2)$$

which reconciles our two formulae without any algebra.

³³By solving for u' :

$$\gamma_{u'} u' = \sqrt{\frac{1}{1/u'^2 - 1/c^2}},$$

so square both sides, take reciprocals, add $1/c^2$, take reciprocals and (magic) take square roots.

Dividing equations (6.40) and (6.39) gives a nice result for the angle α' that the 3-velocity vector $\mathbf{u}'$ in S' makes with the x' axis:

$$\tan \alpha' = \frac{u \sin \alpha}{\gamma(u \cos \alpha - v)}.$$

We can do a similar calculation for the momentum 4-vector of a light ray (a photon). In this case, $|\mathbf{p}| = E/c$, so we can write the components of the 4-momentum vector in the form

$$P = \begin{pmatrix} p \\ p \cos \alpha \\ p \sin \alpha \\ 0 \end{pmatrix}$$

where $p = |\mathbf{p}|$.

Using the Lorentz transformation as before gives

$$\begin{pmatrix} p' \\ p' \cos \alpha' \\ p' \sin \alpha' \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma v/c & 0 & 0 \\ -\gamma v/c & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} p \\ p \cos \alpha \\ p \sin \alpha \\ 0 \end{pmatrix},$$

The t' component gives immediately a formula for the momentum (or energy) of the photon³⁴:

$$E' = p'c = \gamma pc \left(1 - \frac{v}{c} \cos \alpha\right) = \gamma E \left(1 - \frac{v}{c} \cos \alpha\right).$$

Dividing the y' and x' components gives an expression for α' :

$$\tan \alpha' = \frac{\sin \alpha}{\gamma(\cos \alpha - v/c)}$$

or, after some uninteresting algebra,³⁵

$$\tan(\alpha'/2) = \frac{\sqrt{1-v/c}}{\sqrt{1+v/c}} \tan(\alpha/2) = e^\beta \tan(\alpha/2),$$

where $\tanh \beta = v/c$. This is the stellar aberration formula: it gives the change in angle (which corresponds to a change in observed position) of light from stars due to the motion of the Earth.³⁶

6.9 Conservation of 4-momentum

6.9.1 Newton's second law in Special Relativity

The principle of Special Relativity states that the laws of physics hold in all inertial frames. This implies that they must be expressible in terms of invariants and 4-vectors (or 4-tensors). Accordingly, we take Newton's second law for a single relativistic particle to be

$$\frac{dP}{d\tau} = F \tag{6.41}$$

where F is an appropriately defined 4-vector force. We will here always take $F = 0$ so that 4-momentum is conserved. Following the method of Chapter 5, we can extend this to systems of particles, for which the total momentum (i.e. the sum of the momenta of the individual particles) is conserved.

It is beyond the scope of this course to investigate possible right hand sides for the relativistic version of Newton's second law (6.41). The best non-relativistic example was the inverse square gravitational force. Unfortunately, this will not do for relativity: the distance between two particles is invariant under Galilean transformations but is certainly not invariant under Lorentz transformations (because of length contraction). In order to advance to a relativistic theory of gravitation, one has to go to General Relativity, which is exactly that.

³⁴This identical formula appeared in Einstein's 1905 paper 'On the Electrodynamics of Moving Bodies'.

³⁵Note to self: possible STEP question?

³⁶it is a similar effect to the change of angle of rain falling on the windows of a moving train. Generally aberration is the apparent change of an object's direction of movement relative to an observer's frame of reference.

We have mentioned a number of times that Maxwell's equations are invariant under Lorentz transformations so the Lorentz force ought to be a good example. And indeed it is, but it has to be expressed in terms of a second rank field tensor, which again takes us beyond what is possible in this course.³⁷

We will study below the use of momentum conservation in elementary particle reactions. There are two strategies. For both, the first thing to do is to write down the 4-momenta of all the particles involved in a suitable frame, which might be the rest frame of one of the particles or might be the centre of momentum frame, which is the frame in which the total (relativistic) 3-momentum is zero. The first strategy is to obtain equations by conserving the time-like and space-like components of the 4-momenta (i.e. by equating total relativistic energy and momentum). For this strategy, it will usually be necessary to eliminate energy or momentum using (6.33)

$$E^2 = p^2 c^2 + m^2 c^4.$$

The second strategy is to get on one side of the 4-momentum conservation equation one 4-momentum about which one knows nothing and about which one wishes to find nothing and eliminate it using

$$P \cdot P = m^2 c^2.$$

6.9.2 Centre of momentum frame

Sometimes it is helpful to work in the *centre of momentum frame* which is the frame in which the total momentum is zero. That such a frame always exists is not completely obvious. We need some preliminary results.³⁸

Lemma If P_1 and P_2 are timelike or null, and future-pointing³⁹, then $P_1 \cdot P_2 \geq 0$.

Proof If P_1 and P_2 are both null, then they can be written in the form $P_1 = (p_1, \mathbf{p}_1)$ and $P_2 = (p_2, \mathbf{p}_2)$ with $p_i = |\mathbf{p}_i| \geq 0$. Then

$$P_1 \cdot P_2 = p_1 p_2 - \mathbf{p}_1 \cdot \mathbf{p}_2 = p_1 p_2 (1 - \cos \theta) \geq 0,$$

where θ is the angle between the two three-momenta.

If P_1 is timelike, we work in the frame in which $P_1 = (E_1/c, \mathbf{0})$. (We can always think of a timelike 4-vector as the 4-momentum of a particle and this frame is the rest frame of the particle.) Then setting $P_2 = (E_2/c, \mathbf{p}_2)$, we have

$$P_1 \cdot P_2 = E_1 E_2 / c^2 \geq 0.$$

Lemma If P_1 and P_2 are timelike or null and future pointing, then $P_1 + P_2$ is also timelike or null and future pointing.

Proof To demonstrate this, we need to show that $(P_1 + P_2) \cdot (P_1 + P_2) \geq 0$. But

$$\begin{aligned} (P_1 + P_2) \cdot (P_1 + P_2) &= P_1 \cdot P_1 + P_2 \cdot P_2 + 2P_1 \cdot P_2 \\ &\geq 2P_1 \cdot P_2 && \text{(since } P_1 \text{ and } P_2 \text{ are timelike or null)} \\ &\geq 0. && \text{(by the above lemma)} \end{aligned}$$

³⁷Here it is for completeness. We construct the second rank (antisymmetric) 4-tensor from the electric field 3-vector $\mathbf{E}$ and magnetic field 3-vector $\mathbf{B}$ as follows:

$$\mathcal{F} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

It can be checked that this transforms as a 4-tensor should under Lorentz transformations (but you have to know how $\mathbf{E}$ and $\mathbf{B}$ transform). The Lorentz 4-force on a particle with charge q is then

$$q\mathcal{F}U \equiv q\gamma \begin{pmatrix} \mathbf{u} \cdot \mathbf{E} \\ \mathbf{E} + \mathbf{u} \times \mathbf{B} \end{pmatrix},$$

where U is the 4-velocity.

³⁸Note to self: the algebra of 4-vectors with the Minkowski inner product could form a separate expanded section. If P_1 and P_2 are spacelike or null, is it the case that $P_1 + P_2$ is spacelike or null?? (There are obvious counterexamples.) This could be tied in with the action of the (proper) Lorentz group on 4-vectors.

³⁹This means that the first component of each vector is positive

By the above lemmas, the sum of two, and hence any number of, 4-momenta can always be thought of as a 4-momentum (it is timelike and future pointing). Therefore, we can always choose a frame in which the total momentum is zero by transforming to the rest frame of the total momentum.

Another useful result is the following:

Lemma If P_1 and P_2 are 4-momenta of particles with masses m_1 and m_2 respectively (where either might be massless), then $P_1 \cdot P_2 \geq m_1 m_2 c^2$.

Proof If both particles are massless, the result is trivial because the right hand side is zero and the left hand side is non-negative by the lemma on the previous page. Otherwise, we work in the rest frame of one particle:

$$P_1 \cdot P_2 = (m_1 c, \mathbf{0}) \cdot (E_2/c, \mathbf{p}_2) = m_1 E_2 \geq m_1 m_2 c^2,$$

since $E_2 = \sqrt{(\mathbf{p}_2 \cdot \mathbf{p}_2 c^2 + m_2^2 c^4)} \geq m_2 c^2$.

6.9.3 Examples

(i) Decay of a single particle (radioactive decay)

We consider the case of a particle of rest mass m_1 decaying into two particles of rest masses m_2 and m_3 . Conservation of 4-momentum gives (in the obvious notation)

$$P_1 = P_2 + P_3.$$

Thus

$$E_1 = E_2 + E_3 \quad \text{and} \quad \mathbf{p}_1 = \mathbf{p}_2 + \mathbf{p}_3$$

For any particle, the formula $E^2 = p^2 c^2 + m^2 c^4$ holds so

$$E_1 = \sqrt{p_2^2 c^2 + m_2^2 c^4} + \sqrt{p_3^2 c^2 + m_3^2 c^4} \geq m_2 c^2 + m_3 c^2.$$

In particular, considering this result in the rest frame of the decaying particle (where $p_1 = 0$ and $E_1 = m_1 c^2$) gives $m_1 \geq m_2 + m_3$, so (perhaps unsurprisingly) the particle can only decay if its mass is at least as great as the sum of the rest masses of the product particles.

(ii) Decay of a massless particle

We consider the case of a massless particle decaying into two particles of masses m_2 and m_3 (either or both of which might be zero). Again we have

$$P_1 = P_2 + P_3.$$

Taking lengths squared of both sides, and remembering that $P \cdot P = 0$ for a massless particle, we have

$$0 = m_2^2 c^2 + m_3^2 c^2 + 2P_2 \cdot P_3$$

Now $P_2 \cdot P_3 \geq 0$ (see the lemma in the previous section). Since all the terms in this equation are non-negative, they must all vanish. This shows that a massless particle can only decay into massless particles and, if it does decay to two massless particles, their 3-momenta must be parallel.

(iii) Particle creation

If a proton⁴⁰ collides at sufficiently high energy with another proton, a proton anti-proton pair can be created, in addition to the original protons:⁴¹

$$p + p \rightarrow p + p + (p + \bar{p}).$$

We define the kinetic energy of a particle to be its energy minus its rest energy, i.e. $m\gamma c^2 - mc^2$; this is of course frame dependent. One might have thought that the minimum kinetic energy necessary to produce the extra two protons would be just $2m_p c^2$, but it turns out that much more is required. In fact, we shall see that the minimum energy required is $6m_p c^2$.

⁴⁰Protons are denoted by p , anti-protons by $\bar{p}$; protons and anti-protons have the same rest mass m_p .

⁴¹The first production of anti-protons (on Earth) was achieved by Chamberlain and Segre at the Berkley Bevatron in California; they received the Nobel prize for this in 1959.

By conservation of 4-momentum we have

$$P = Q_1 + Q_2 + Q_3 + Q_4$$

where P is the total 4-momentum before collision (the sum of the individual 4-momenta of the two incident protons) and Q_i are the 4-momenta after the collision of the two original protons and the created proton and anti-proton. If we work in the rest frame of one of the two incident protons, so that $P = (m_p c, 0) + (E/c, \mathbf{p})$, we see that

$$P \cdot P = m_p^2 c^2 + m_p^2 c^2 + 2(m_p c)(E/c) = 2m_p^2 c^2 + 2m_p E.$$

We have

$$P \cdot P = (Q_1 + Q_2 + Q_3 + Q_4)(Q_1 + Q_2 + Q_3 + Q_4)$$

and, because of the lemma in the previous section, the magnitude of each of the 16 terms in the expansion of the right hand side of this equation is at least $m_p^2 c^2$. Thus

$$2m_p^2 c^2 + 2m_p E \geq 16m_p^2 c^2 \quad \text{i.e.} \quad E - m_p c^2 \geq 6m_p c^2$$

which is the minimum kinetic energy required to form the proton anti-proton pair. This reflects the fact that there must be enough incident energy (at least, in the case considered here of an incident proton colliding with a stationary proton) to account for the recoil kinetic energy of all the particles as well as the rest masses of the created particles. Of course, there is no Newtonian analogue or Newtonian limit for this particular example, because in this example rest mass is not conserved.⁴²

(iv) The same as (iii) but using the centre of momentum frame

In the rest frame of one of the incident protons, the total momentum of the two incident protons is $(m_p c, 0) + (E/c, \mathbf{p}) = (m_p c + E/c, \mathbf{p})$ as above.

Suppose now that we transform (without actually doing the transformation) to the centre of momentum frame, in which the total momentum is zero. In this frame, the total momentum of the incident protons is $(E'/c, 0)$ where E' is the (not yet calculated) total relativistic energy. Thus the total relativistic energy of the four protons after the collision, $E_1 + E_2 + E_3 + E_4$ satisfies

$$E' = E_1 + E_2 + E_3 + E_4 \geq 4m_p c^2.$$

Furthermore, this bound cannot be improved because the inequality becomes an equality if all four protons have zero 3-momentum.

To find out what this inequality says about E , we have to transform back into the original frame. Let us choose axes so that $\mathbf{p}$ is parallel to the x -axis. Then, using our standard Lorentz transformation,

$$\begin{pmatrix} E'/c \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma & \gamma v/c \\ \gamma v/c & \gamma \end{pmatrix} \begin{pmatrix} m_p c + E/c \\ p \end{pmatrix}$$

The second row of this matrix equation tells us the velocity of the transformation and the first row will tell us, using the velocity, we have just found, what E' is in terms of $E + m_p c^2$ and p . But hold on! Do we actually have to do this rather messy transformation? Could we not just use the scalars?

Indeed we can. The scalar magnitude of the 4-momentum is of course the same in both (all) frames, so

$$E'^2/c^2 = (m_p c + E/c)^2 - p^2$$

and

$$(m_p c + E/c)^2 - p^2 \geq 16m_p^2 c^2.$$

Now expanding the square on the left hand side, and remembering that $E^2 = p^2 c^2 + m_p^2 c^4$, we find that

$$2m_p E \geq 14m_p^2 c^2,$$

as before.

But hold on! Surely if we are using 4-vectors, we should be able to avoid using results such as $E^2 = p^2 c^2 + m_p^2 c^4$, because this can be derived using the invariance of $P \cdot P$? And indeed we can. Writing the incoming proton momenta as P_1 and P_2 , we have for the total momentum before the collision

$$(P_1 + P_2) \cdot (P_1 + P_2) = P_1 \cdot P_1 + P_2 \cdot P_2 + 2P_1 \cdot P_2 = 2m_p^2 c^2 + 2P_1 \cdot P_2.$$

⁴²Note to self: this example could be extended by repeating it in the centre of momentum frame and relating the results by a Lorentz transformation.

In the original frame $P_1 = (m_p c, \mathbf{0})$ and $P_2 = (E/c, \mathbf{p})$, so

$$2m_p^2 + 2Em_p = (Q_1 + Q_2 + Q_3 + Q_4)(Q_1 + Q_2 + Q_3 + Q_4) \geq 16m_p^2 c^2,$$

giving the same result as before.

By working in the centre of momentum, we have established this result rather easily, and we have also proved that no tighter bound is possible.

(v) Pion decay

We consider the decay of the neutral pion⁴³ into two photons:

$$\pi^0 \rightarrow \gamma + \gamma$$

Suppose the energy of the pion (in some given frame) is E_π and the energy of one of the photons (in the same frame) E_γ . What is the angle, θ , between the path of this photon and the path of the pion?

By conservation of 4-momentum we have, in the obvious notation,

$$P_\pi = P_\gamma + P'_\gamma$$

We know nothing about the 4-momentum P'_γ of the second photon, except that it is a null (light-like) vector and therefore has zero length. Nor are we asked to find out anything about it. We therefore eliminate it by taking lengths. We have

$$P'_\gamma = P_\pi - P_\gamma$$

so

$$\begin{aligned} 0 &= (P_\pi - P_\gamma) \cdot (P_\pi - P_\gamma) = P_\pi \cdot P_\pi + P_\gamma \cdot P_\gamma - 2P_\pi \cdot P_\gamma \\ &= m_\pi^2 c^2 + 0 - 2E_\pi E_\gamma / c^2 + 2\mathbf{p}_\pi \cdot \mathbf{p}_\gamma \\ &= m_\pi^2 c^2 - 2E_\pi E_\gamma / c^2 + 2 \left(\sqrt{E_\pi^2 / c^2 - m_\pi^2 c^2} \right) E_\gamma \cos \theta \end{aligned}$$

(recall that for a photon $E = pc$). In this last equation, everything except θ is known, so θ can be found in terms of the known quantities.

⁴³There is a triplet of pions with positive, negative and zero charge ($\pi^\pm$ and π^0). They are mesons, i.e. middle weight particles compared with hadrons (heavy particles such as neutrons and protons) and leptons (light particles such as electrons). The neutral pion has a short life-time: it decays with a half life of 10^{-17} seconds, mostly by the process described in this example.